Slicing Through the 3D World 🍰🧠

A lightning-fast software engine that turns massive 3D objects into beautiful 2D movies.

🔗 github.com/moatasem/OOP-3D-Projection

The "Virtual Knife" Concept

> **Imagine having a magical, transparent knife.** You are handed a massive 3D object—like a medical MRI scan of a human brain, or a 3D scan of a delicious croissant. Instead of just looking at the outside, you want to know what it looks like on the inside, from absolutely any angle. This software acts as that virtual knife, slicing through digital objects to show you exactly what is hidden inside.

How Does It Work?

1️⃣

Feed it a 3D Shape

We load a massive 3D computer file. It can be anything from medical data to food scans, made up of millions of tiny blocks called "voxels".

2️⃣

Choose the Angle

Using basic coordinates, we tell the software exactly how to position our virtual knife. You can slice it straight down, sideways, or at a crazy diagonal.

3️⃣

Action! 🎬

The engine rapidly calculates the colors inside the object along that slice, turning the inside of the 3D object into a clear 2D movie!

What Can We Slice?

The project was successfully tested on a diverse range of 3D objects. It isn't just for boring math—it handles complex, real-world shapes!

🧠 Human Brain (MRI) | 🥐 A French Croissant | 🍰 Swiss Roll Cake | 🧅 Layered Onion | 🍪 Biscuit Cake

Mind-Blowing Speed 🚀

Calculating exactly what is inside a 3D object requires a massive amount of math. Normally, processing an object with **125 million data points** is slow and laggy. By using advanced software engineering tricks (writing Python code that compiles directly into C-level machine code), the engine breaks all speed limits.

30 FPS	125M	1,623+ FPS
MINIMUM TARGET GOAL	DATA POINTS (VOXELS)	OUR ACTUAL SPEED